Part A – Theory

Introduction to Unsupervised Learning and Clustering

Introduction

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make decisions without being explicitly programmed. Machine learning techniques are generally divided into supervised learning, unsupervised learning, and reinforcement learning. In this assignment, the main focus is on unsupervised learning and clustering, which are commonly used to discover hidden patterns in data.

What is Unsupervised Learning?

Unsupervised learning is a machine learning approach in which the algorithm learns from data that does not have predefined labels or target values. Instead of predicting a known outcome, the model explores the dataset to identify similarities, differences, and hidden structures among the observations.

The main objective of unsupervised learning is to organize data into meaningful groups or discover relationships that may not be obvious. One of the most common tasks in unsupervised learning is clustering, where similar data points are grouped together based on their characteristics.

For example, a wholesale company can use customer purchasing data to automatically group customers with similar buying habits. These groups can then be used for marketing, inventory planning, and customer relationship management.

Difference Between Unsupervised Learning, Regression, and Classification

Although all three methods are part of machine learning, they solve different types of problems.

- **Unsupervised Learning:** Works with unlabeled data and discovers hidden patterns or groups without knowing the correct answers beforehand.
- **Regression:** A supervised learning technique that predicts continuous numerical values such as house prices, salaries, or temperatures.
- **Classification:** A supervised learning technique that predicts categorical labels such as "Approved" or "Rejected", "Spam" or "Not Spam", and "Disease" or "No Disease".

The key difference is that supervised learning requires labeled training data, while unsupervised learning works without labels and focuses on finding the natural structure of the data.

Real-Life Examples

Example of Clustering

A supermarket or wholesale distributor can analyze customers' purchasing behavior and group customers with similar spending patterns. One cluster may contain customers who purchase large amounts of fresh products, while another cluster may mainly purchase grocery and cleaning products. These customer segments help businesses create targeted marketing campaigns and improve inventory management.

Example of Supervised Learning

A bank can use supervised learning to predict whether a loan application should be approved or rejected. The model is trained using historical loan records where each application already has a known outcome. After training, the model can classify new applicants based on their financial information.